A set is a 1-type

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 117-120 and its proof environment opens on line 121, the next line; the pairing is by adjacency and the retained proof range 121-128 is the whole environment.

Editorial note. The retained counter values (chapter 3, section 1, theorem offset 7) make the counted lemma print as Lemma 3.1.8, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:isset-is1type = 3.1.8).

Editorial note. Two labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: cor:transport-path-prepost (book Corollary 2.11.2, whose chapter file is not part of the pinned corpus so the pointer span is the label line of this chapter dump) and defn:1type (book Definition 3.1.7, defined at line 101 of this chapter pinned file).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: a set is a 1-type (author label thm:isset-is1type, book Lemma 3.1.8).

Allowed standard prerequisites (editorial).

- Set and 1-type. A set is a type whose path types are mere propositions, and a 1-type is a type whose path types are sets.. They are the two levels related by the statement.
- Dependent action on paths and transport in a path family. The action on paths of a dependent function yields a dependent path, and transport in a family of paths computes by concatenation.. They turn the action on paths into an explicit concatenation identity.
- Cancellation. A common left factor of an equality can be cancelled.. It finishes the argument.

Source reference. Original corollary 2.11.2: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 188–188. Book Corollary 2.11.2, the computation of transport in a family of paths by concatenation, used at source line 125. Navigation pointer only: the chapter file that carries it is not part of the pinned corpus, so the pointer span is the label line of this chapter dump.

Source reference. Original definition 3.1.7: sources/hott-logic/pinned-original-source.tex, source lines 101–101. Book Definition 3.1.7, the definition of a 1-type, recalled at source line 127. Navigation pointer only.

Lemma 3.1.8. *If A is a set (that is, $\mathsf{isSet}(A)$ is inhabited), then A is a 1-type.*

Proof. Suppose $f : \mathsf{isSet}(A)$; then for any $x, y : A$ and $p, q : x = y$ we have $f(x, y, p, q) : p = q$. Fix x, y, and p, and define $g : \prod_{(q:x=y)}(p = q)$ by $g(q) :\equiv f(x, y, p, q)$. Then for any $r : q = q'$, we have $\mathsf{apd}_g(r) : r_*(g(q)) = g(q')$. By original corollary 2.11.2, therefore, we have $g(q) \cdot r = g(q')$.

In particular, suppose given x, y, p, q and $r, s : p = q$, as in original definition 3.1.7, and define g as above. Then $g(p) \cdot r = g(q)$ and also $g(p) \cdot s = g(q)$, hence by cancellation $r = s$. □